21.	Answer: **B** $Q = It = \frac{V}{R}t$ When all 3 switches are closed, $\mathrm{R}_{eff} = \left(\frac{1}{4.0} + \frac{1}{6.0}\right)^{-1} = 2.4\ \Omega$ (note that a short- $Q_{\text{first minute}} = \frac{V}{\mathrm{R}_{eff}}t = \frac{10}{2.4}(60)$ When S2 and S3 are opened, $R_{\text{eff}} = 6.0 + 2.0 = 8.0\ \Omega$ $Q_{\text{second minute}} = \frac{V}{\mathrm{R}_{eff}}t = \frac{10}{8.0}(60)$ $Q_{\text{total}} = \frac{10}{2.4}(60) + \frac{10}{8.0}(60) = 325\ \mathrm{C}$